Assignment – 12

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Question 1

Method Overriding – Payment System (10 Marks). Create a parent class named .
Implement a method: makePayment()
Create the following child classes:

- **CreditCardPayment**
- **UPIPayment**
- **CashPayment**

Each child class should the makePayment() method and display an appropriate payment message.
Requirements:

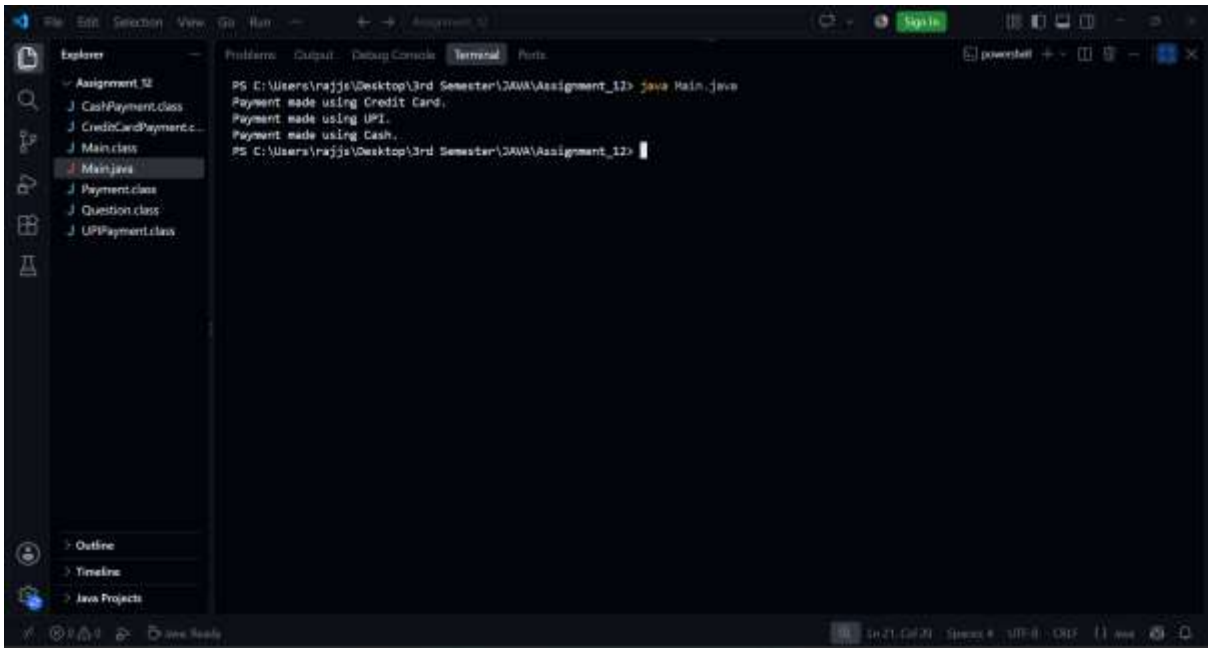
- **Implement .**
- **Create objects of all child classes.**
- **Call the makePayment() method for each object.**
- **Demonstrate using a parent class reference.**

Answer-

```
class Payment {
   void makePayment() {
      System.out.println("Making a payment...");
   }
}
class CreditCardPayment extends Payment {
   void makePayment() {
      System.out.println("Payment made using Credit Card.");
   }
}
class UPIPayment extends Payment {
   void makePayment() {
      System.out.println("Payment made using UPI.");
   }
}
```

```
class CashPayment extends Payment {
    void makePayment() {
        System.out.println("Payment made using Cash.");
    }
}
public class Main {
    public static void main(String[] args) {
        Payment p;
        p = new CreditCardPayment();
        p.makePayment();
        p = new UPIPayment();
        p.makePayment();
        p = new CashPayment();
        p.makePayment();
    }
}
```

OUTPUT-



Question 2

Runtime Polymorphism – Employee Salary System (10 Marks)
Create a parent class named .
Declare:

- **Employee ID**
- **Employee Name**

Create the following child classes:

- **FullTimeEmployee**
- **PartTimeEmployee**

Implement a method: calculateSalary() Requirements:

- **Calculate salary based on a fixed monthly salary.**
- **Calculate salary based on:**
- **Hours Worked**
- **Payment Per Hour**

Demonstrate:

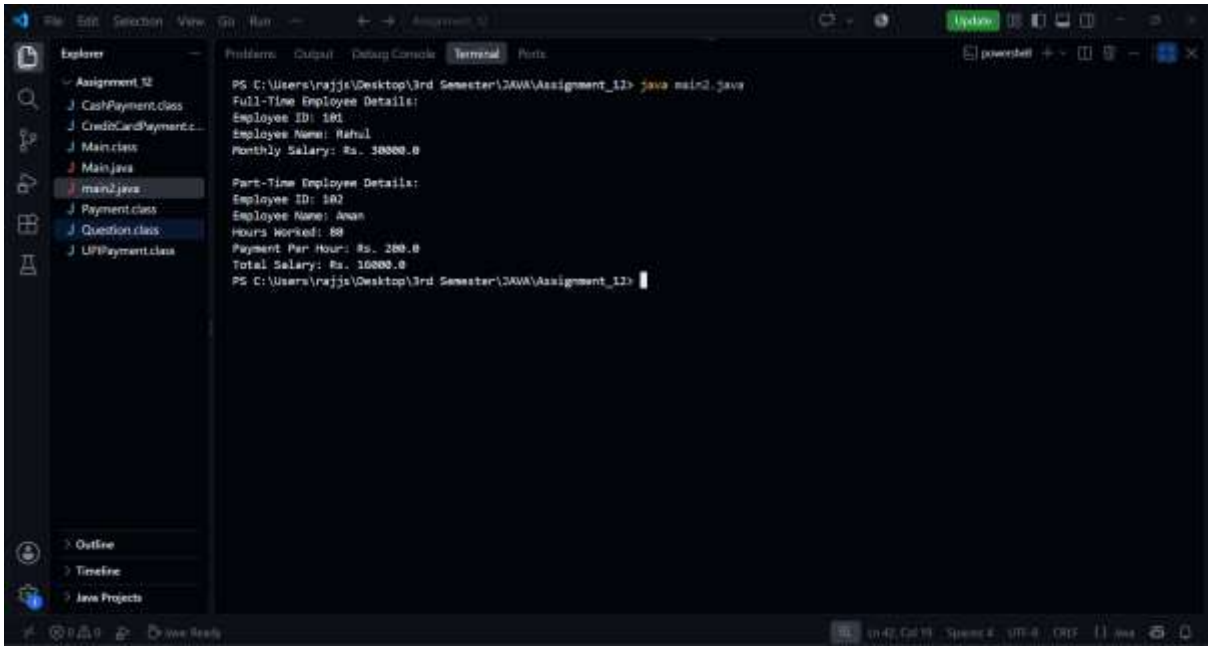
- **Method overriding.**
- **Runtime polymorphism.**
- **Calling the calculateSalary() method using a parent class reference.**

Answer-

```
class Employee {
   int employeeId;
   String employeeName;
   Employee(int employeeId, String employeeName) {
      this.employeeId = employeeId;
      this.employeeName = employeeName;
   }
   void calculateSalary() {
      System.out.println("Calculating employee salary...");
   }
}
class FullTimeEmployee extends Employee {
   double monthlySalary;
   FullTimeEmployee(int employeeId, String employeeName, double monthlySalary) {
```

```
        super(employeeId, employeeName);
        this.monthlySalary = monthlySalary;
    }
    void calculateSalary() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Employee Name: " + employeeName);
        System.out.println("Monthly Salary: Rs. " + monthlySalary);
    }
}
class PartTimeEmployee extends Employee {
    int hoursWorked;
    double paymentPerHour;
    PartTimeEmployee(int employeeId, String employeeName,
                int hoursWorked, double paymentPerHour) {
        super(employeeId, employeeName);
        this.hoursWorked = hoursWorked;
        this.paymentPerHour = paymentPerHour;
    }
    void calculateSalary() {
        double salary = hoursWorked * paymentPerHour;
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Employee Name: " + employeeName);
        System.out.println("Hours Worked: " + hoursWorked);
        System.out.println("Payment Per Hour: Rs. " + paymentPerHour);
        System.out.println("Total Salary: Rs. " + salary);
    }
}
public class main2 {
    public static void main(String[] args) {
        Employee e;
        e = new FullTimeEmployee(101, "Rahul", 30000);
        System.out.println("Full-Time Employee Details:");
        e.calculateSalary();
        System.out.println();
        e = new PartTimeEmployee(102, "Aman", 80, 200);
        System.out.println("Part-Time Employee Details:");
        e.calculateSalary();
    }
}
```

OUTPUT-


Explorer
Assignment 12
CashPayment.class
CreditCardPayment.c...
Main.class
Main.java
main2.java
Payment.class
Question.class
UPIPayment.class
Outline
Timeline
Java Projects
Problems
Output
Debug Console
Terminal
Ports
powershell
PS C:\Users\rajjs\Desktop\3rd Semester\JAVA\Assignment_12> java main2.java
Full-Time Employee Details:
Employee ID: 101
Employee Name: Rahul
Monthly Salary: Rs. 30000.0
Part-Time Employee Details:
Employee ID: 102
Employee Name: Aman
Hours Worked: 80
Payment Per Hour: Rs. 200.0
Total Salary: Rs. 16000.0
PS C:\Users\rajjs\Desktop\3rd Semester\JAVA\Assignment_12>